

1. ASCENDING ORDER

Flag: S 0, Z 1, AC 0, P 1, C 0

Load me at: []

version: Hex, To Dec, value, ony

```
1 LDA 2500
2 DCR A
3 MOV B,A
4
5 OUTER: LXI H,2501
6 MOV C,B
7
8 INNER: MOV A,M
9 MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JC NOSWAP
16 JZ NOSWAP
17
18 MOV M,D
19 DCR H
20 MOV M,E
21 INX H
22
23 NOSWAP: DCR C
24 JNZ INNER
25 DCR B
26 JNZ OUTER
27
28 HLT
```

Start: 2500

Address (Hex)	Address	Data
09C4	2500	4
09C5	2501	1
09C6	2502	2
09C7	2503	5
09C8	2504	8
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No: 0, Assembler Message: Program assembled successfully

2. DESCENDING ORDER

Start: 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	8
09C6	2502	8
09C7	2503	2
09C8	2504	1
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No: 0, Assembler Message: Program assembled successfully

3. LARGEST NUMBER

Flag

S 0

Z 1

AC 0

P 1

C 1

Inversion

Hex

◀ To Dec

Value

E

Memory

Load me at

```

1  LXI  H,2501
2  MOV  B,M
3  MOV  A,M
4  MOV  D,A
5  DCR  B
6
7  LOOP: INX H
8  MOV  A,M
9  CMP  D
10 JC  SKIP
11 MOV  D,A
12
13 SKIP: DCR B
14 JNZ  LOOP
15
16 MOV  A,D
17 STA 2506
18 HLT

```

Data

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	45
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Stack

KeyPad

Memory

Line No Assembler Message

0 Program assembled successfully

4. MINIMUM NUMBER

5

Load me at

```

1  LXI H,2501
2  MOV B,M
3  MOV A,M
4  MOV D,A
5  DCR B
6
7  LOOP: INX H
8  MOV A,M
9  CMP D
10 JNC SKIP
11 MOV D,A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A,D
17 STA 2600
18 HLT

```

Data
Stack
KeyPad
Memory

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	5
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No	Assembler Message
0	Program assembled successfully

5. LCM OF 2 NUMBERS

Flag

S 0

Z 1

DD AC 0

P 1

FF C 0

Load me at

1 LDA 2500

2 MOV B,A

3 LDA 2501

4 MOV C,A

5

6 MOV D,B

7 MOV E,C

8

9 LOOP: MOV A,D

10 CMP E

11 JZ DONE

12 JC ADDD

13

14 MOV A,E

15 ADD C

16 MOV E,A

17 JMP LOOP

18

19 ADDD: MOV A,D

20 ADD B

21 MOV D,A

22 JMP LOOP

23

24 DONE: MOV A,D

25 STA 2502

26 HLT

Conversion

Hex

0

To Dec

00

Port Value

18

Memory

Data Stack KeyPad Memory I/O Ports

Start 2500 OK

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	24
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0 Program assembled successfully

6. GCD OF 2 NUMBERS

Flag

S 0

Z 1

DD AC 0

P 1

FF C 0

Load me at

1 LDA 2500

2 MOV B,A

3 LDA 2501

4 MOV C,A

5

6 LOOP: MOV A,B

7 CMP C

8 JZ DONE

9 JC SECOND

10 SUB C

11 MOV B,A

12 JMP LOOP

13

14 SECOND: MOV A,C

15 SUB B

16 MOV C,A

17 JMP LOOP

18

19 DONE: MOV A,B

20 STA 2502

21 HLT

I - Hex Conversion

Hex

0

To Dec

00

Update Port Value

02

Update Memory

Data Stack KeyPad Memory I/O Ports

Start 2500 OK

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0 Program assembled successfully

7. FACTORIAL OF A NUMBER

Flag

S 0

Z 1

AC 0

P 1

C 0

Conversion

Hex

0

To Dec

00

Port Value

05

Memory

Load me at

1 LDA 2500

2 MOV C, A

3 MVI B, 01H

4

5 FACT: MOV A, C

6 CPI 00H

7 JZ DONE

8

9 MOV D, C

10 MOV A, B

11 MOV E, A

12 MVI A, 00H

13

14 MUL: ADD E

15 DCR D

16 JNZ MUL

17

18 MOV B, A

19 DCR C

20 JMP FACT

21

22 DONE: MOV A, B

23 STA 2501

24 HLT

Data

Stack

Keypad

Me

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	120
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0

Program assembled successfully

8. DECIMAL TO HEXADECIMAL

S 0

Z 0

AC 1

P 1

C 0

Conversion

Hex

To Dec

Value

Memory

1 LDA 2500

2 MOV B, A

3 ANI 0FH

4 MOV E, A

5

6 MOV A, B

7 ANI 0F0H

8 RRC

9 RRC

10 RRC

11 RRC

12 MOV D, A

13

14 MVI C, 0AH

15 MVI A, 00H

16

17 LOOP: ADD D

18 DCR C

19 JNZ LOOP

20

21 ADD E

22 STA 2501

23 HLT

Data

Stack

Keypad

Me

Start 2500

Address (Hex)	Address	Data
09C4	2500	45
09C5	2501	33
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0

Program assembled successfully